

Benhur Okur

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PROFILE

Computer Engineering graduate with hands-on experience in software development, artificial intelligence, and data science. Builds full-stack and systems-oriented projects with a focus on secure architecture, practical problem solving, and production-minded design. Interested in software engineering roles involving backend systems, AI-powered products, and scalable application development.

EDUCATION

Izmir University of Economics

Sep. 2022 – Jun. 2026

Bachelor of Science in Computer Engineering

Izmir, Türkiye

- Full Scholarship recipient
- Graduated: June 2026
- Relevant Coursework: Programming Fundamentals, Data Structures and Algorithms, Database Systems, Operating Systems, Software Engineering, Programming Paradigms, Computer Networks, Data Science, Artificial Intelligence, Machine Learning, Digital Image Processing

EXPERIENCE

SOCRadar Cyber Intelligence

Jul. 2025 – Aug. 2025

Software Engineer Intern

Istanbul, Türkiye

- Developed a full-stack LLM Proxy Service that unified access to multiple AI providers through a single platform.
- Implemented OAuth2 authentication and JWT-based session flows to improve account security and access control.
- Built encrypted API key handling to protect sensitive credentials used in external model integrations.
- Designed a modular routing structure that simplified model selection and improved extensibility for future providers.
- Created a responsive dashboard with React and Tailwind CSS to streamline API management and user interaction.
- Tested and debugged core workflows to improve reliability and production readiness.

Freelance Systems Developer

Jul. 2024 – Aug. 2025

Self-Employed

Remote

- Worked as an independent freelance developer delivering low-level systems software for clients, with a primary focus on the DaemonSystems messaging daemon project.
- Designed and implemented a UNIX-based multithreaded messaging daemon in C for concurrent message processing.
- Built a thread-safe circular buffer architecture to ensure stable message dispatch under parallel workloads.
- Implemented filtering, file I/O, and inter-process communication mechanisms to improve message handling efficiency.
- Strengthened system reliability through low-level concurrency-focused development practices.

PROJECTS

AutoEDA — Human-in-the-Loop Data Cleaning Platform | *FastAPI, Celery, PostgreSQL, Redis, LLM, Senior Capstone*

- Architected a web platform that automates data quality analysis and cleaning while keeping a human in control — every cleaning action requires explicit user approval and the original dataset is never mutated, a trust-and-safety guarantee missing from existing one-click AutoML and data-prep tools.
- Engineered a hybrid detection engine that pairs deterministic, rule-based issue detection with LLM-generated plain-English explanations, ensuring the AI explains issues but never silently decides cleaning actions — solving the “black-box automation” problem that blocks adoption in regulated, audit-sensitive environments.
- Built an asynchronous execution pipeline with FastAPI, Celery, and Redis that applies approved pandas transformations without blocking, returning task IDs instantly and generating downloadable HTML data-quality reports.
- Modeled the full dataset lifecycle (upload → profiling → approval → cleaning) in PostgreSQL with async SQLAlchemy, tracking the state of every suggestion for end-to-end traceability and reproducibility.

LLM Proxy Service | *Flask, React, OAuth2, JWT, REST APIs*

- Designed and developed a modular LLM proxy platform that unified access to multiple AI providers through a single interface.

- Reduced integration complexity by centralizing authentication, model routing, and API key management in one service layer.
- Implemented secure credential handling and user-based access flows to support safer multi-provider usage.
- Built the system with extensibility in mind, enabling new model providers and backend features to be added with minimal architectural changes.

Breath Detection System | *Python, Librosa, Machine Learning*

- Developing a real-time respiratory activity detection system using audio signal processing and machine learning methods.
- Designed a processing pipeline to extract breathing-related patterns from ambient audio for low-cost monitoring scenarios.
- Targeted practical healthcare-oriented use cases by exploring software-based detection as an alternative to specialized hardware.

Izmir Public Transportation Ridership Analysis | *Python, Pandas, Scikit-learn, Jupyter*

- Analyzed 2015–2022 public transportation ridership data from Izmir (Eshot, IZBAN, Izdeniz) as a semester-long data science team project to uncover usage trends, seasonal patterns, and anomalies.
- Performed exploratory data analysis with visualizations (histograms, line plots, box plots) to characterize ridership behavior across transit operators and fare types.
- Built a preprocessing pipeline handling missing values, normalization/standardization, feature selection, and PCA to prepare the time-series dataset for modeling.
- Applied data mining techniques such as clustering and time-series forecasting to extract deeper insights from daily ridership counts.

BookCatalog | *Java, JavaFX, JSON*

- Developed a desktop-based book management application to organize and manage records more efficiently.
- Implemented full CRUD functionality to simplify everyday data entry, updating, and retrieval workflows.
- Designed a user-friendly JavaFX interface that improved accessibility for non-technical users.

SKILLS

Languages: Python, Java, C, C#, JavaScript, SQL, Kotlin

Frameworks & Libraries: FastAPI, Flask, React, Tailwind CSS, SQLAlchemy, Pandas, Celery, JavaFX

Tools & Technologies: Git, GitHub, Docker, Redis, REST APIs, JWT Authentication, OAuth2, SSE (Server-Sent Events), ThreadPool / Concurrent Programming, Unix/Linux

Databases: PostgreSQL, MySQL, SQLite

Core Areas: Backend Development, Full-Stack Development, API Design, Distributed Systems, Machine Learning, Data Analysis

ADDITIONAL INFORMATION

Languages: Turkish, English, German